

EC-467 Mobile Application Development



BS (CE)-2021 LAB REPORT #8

Submitted By:

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	Presentation (Format(0.5)+ Title (0.5)+ Conclusion(1) +Procedure(2) +Objective(1))	Calculation/ Coding	Observation/ Program Results	Total
Total	5	5	5	15
Obtained				

Lab Instructor.

Dr. Muhammad Bilal

LAB REPORT #8

Fragments

Introduction:

A Fragment is a piece of an activity which enable more modular activity design. A fragment is a kind of sub-activity. A Fragment represents a behavior or a portion of user interface in a Fragment Activity.

Objectives of the experiment:

- 1.To get basic understanding regarding fragments.
- 2.To get basic understanding regarding how fragments pass information between fragments.
- 3.To get to know about the benefits related to Fragments.

Tools:

Android Studio Jellyfish

Class Task:

Fragments

JAVA CODE OF MAIN ACTIVITY:

```
package com.huzaifashahbaz.fragmentsapp;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import androidx.appcompat.app.AppCompatActivity;
import androidx.fragment.app.Fragment;
import androidx.fragment.app.FragmentManager;
import androidx.fragment.app.FragmentTransaction;
public class MainActivity extends AppCompatActivity {
    boolean isFragmentA=true;
    FragmentManager fragmentManager;
    FragmentTransaction fragmentTransaction;
    FragmentA fragmentA;
    FragmentB fragmentB;
```

```

String messageA="Message from Main for Fragment A";
String messageB="Message from Main for Fragment B";
Bundle args=new Bundle();
public static final String MSG_KEY="MSG_KEY";
@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
    fragmentA=new FragmentA();
    fragmentB=new FragmentB();
    replaceFragment(fragmentA,messageA);
    Button replaceBtn=findViewById(R.id.replace_btn);
    replaceBtn.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View view) {
            if (!isFragmentA)
            {
                replaceFragment(fragmentA,messageA);
                isFragmentA=true;
            }
            else
            {
                replaceFragment(fragmentB,messageB);
                isFragmentA=false;
            }
        }
    });
}
void replaceFragment(Fragment fragment, String message)
{
    args.putString(MSG_KEY,message);
    fragment.setArguments(args);
    fragmentManager=getSupportFragmentManager();
    fragmentTransaction=fragmentManager.beginTransaction();
    fragmentTransaction.replace(R.id.frame_container,fragment);
    fragmentTransaction.addToBackStack(null);
    fragmentTransaction.commit();
}
}

```

XML CODE OF MAIN ACTIVITY:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:weightSum="4"
    tools:context=".MainActivity">
    <LinearLayout
        android:orientation="vertical"
        android:id="@+id/frame_container"
        android:layout_weight="3"
        android:layout_width="match_parent"
        android:layout_height="match_parent">
    </LinearLayout>
    <Button
        android:id="@+id/replace_btn"
        android:layout_weight="1"
        android:layout_gravity="center"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Replace" />
</LinearLayout>
```

JAVA CODE OF FRAGMENT-A:

```
package com.huzafashahbaz.fragmentsapp;
import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.TextView;
import androidx.annotation.NonNull;
```

```

import androidx.annotation.Nullable;
import androidx.fragment.app.Fragment;
public class FragmentA extends Fragment {
    @Nullable
    @Override
    public View onCreateView(@NonNull LayoutInflater inflater, @Nullable
    ViewGroup container, @Nullable Bundle savedInstanceState)
    {
        View
        fragmentView=inflater.inflate(R.layout.fragment_a_layout,container,false);
        TextView fragmentTV=fragmentView.findViewById(R.id.main_message_a);
        if (getArguments()!=null)
        {
            String msg=getArguments().getString(MainActivity.MSG_KEY);
            fragmentTV.setText(msg);
        }
        return fragmentView;
    }
}

```

XML CODE OF FRAGMENT-A:

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:orientation="vertical"
    android:padding="@dimen/view_margin"
    android:background="@color/green"
    android:layout_width="match_parent"
    android:weightSum="10"
    android:layout_height="match_parent">
    <TextView
        android:id="@+id/fragment_title_a"
        android:gravity="center"
        android:textSize="32sp"
        android:textColor="@color/white"
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_weight="8"
        android:text="Fragment A" />

```

```

<TextView
    android:id="@+id/main_message_a"
    android:gravity="center"
    android:textColor="@color/white"
    android:layout_width="match_parent"
    android:layout_height="0dp"
    android:layout_weight="2"
    android:text="Message from main" />
</LinearLayout>

```

JAVA CODE OF FRAGMENT-B:

```
package com.huzafashahbaz.fragmentsapp;
```

```

import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.TextView;

```

```

import androidx.annotation.NonNull;
import androidx.annotation.Nullable;
import androidx.fragment.app.Fragment;

```

```

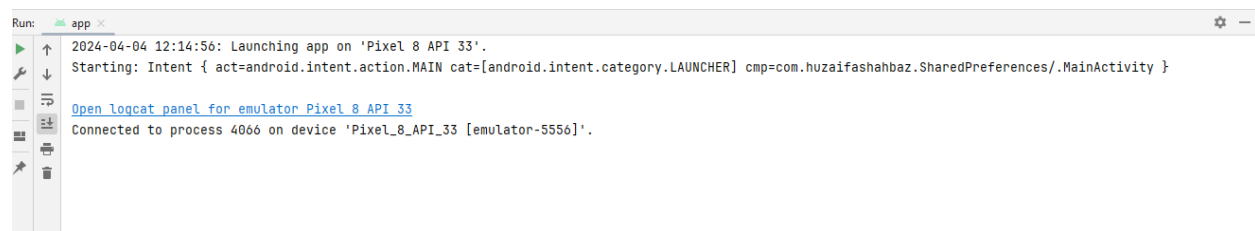
public class FragmentB extends Fragment {
    @Nullable
    @Override
    public View onCreateView(@NonNull LayoutInflater inflater, @Nullable
        ViewGroup container, @Nullable Bundle savedInstanceState)
    {
        View fragmentView=inflater.inflate(R.id.fragment_b_layout,container,false);
        TextView fragmentTV=fragmentView.findViewById(R.id.main_message_b);
        if (getArguments()!=null)
        {
            String msg=getArguments().getString(MainActivity.MSG_KEY);
            fragmentTV.setText(msg);
        }
    }
    return fragmentView;
}

```

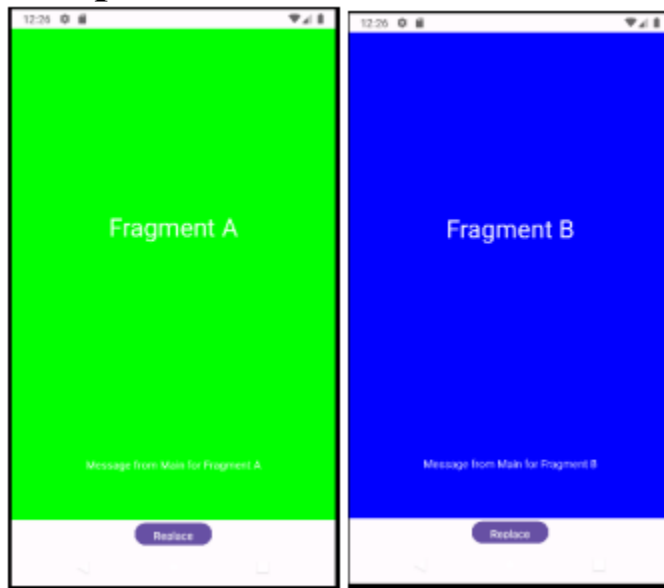
XML CODE OF FRAGMENT-B:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:orientation="vertical"
    android:padding="@dimen/view_margin"
    android:background="@color/blue"
    android:layout_width="match_parent"
    android:weightSum="10"
    android:layout_height="match_parent">
    <TextView
        android:id="@+id/fragment_title_a"
        android:gravity="center"
        android:textSize="32sp"
        android:textColor="@color/white"
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_weight="8"
        android:text="Fragment B" />
    <TextView
        android:id="@+id/main_message_b_tv"
        android:gravity="center"
        android:textColor="@color/white"
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_weight="2"
        android:text="Message from main" />
</LinearLayout>
```

Results:



Output:



Home Task:

Use Tasbeeh app and show history in separate fragment instead of separate activity.

Note: Bundle can share string array list

JAVA CODE OF MAIN ACTIVITY:

```
package com.huzaifashahbaz.tabesshfragmentapp;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import androidx.fragment.app.FragmentManager;
import androidx.fragment.app.FragmentTransaction;
import java.util.ArrayList;
public class MainActivity extends AppCompatActivity {
    TextView tvCount;
    Button btnCount, btnReset, btnViewHistory;
    int Count = 0;
    ArrayList<String> historyList = new ArrayList<>();
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
```



```

setContentView(R.layout.activity_main);
tvCount = findViewById(R.id.text_view_count);
btnCount = findViewById(R.id.btn_count);
btnReset = findViewById(R.id.btn_reset);
btnViewHistory = findViewById(R.id.btn_view_history);
btnCount.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        onPressCountbtn(v);
    }
});
btnReset.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        onPressResetbtn(v);
    }
});
btnViewHistory.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        onPressHistorybtn(v);
    }
});
}
public void onPressResetbtn(View view) {
    Count = 0;
    tvCount.setText("Count: " + Count);
    historyList.clear();
}
public void onPressHistorybtn(View view) {
    FragmentManager fragmentManager = getSupportFragmentManager();
    FragmentTransaction fragmentTransaction =
fragmentManager.beginTransaction();
    HistoryFragment fragment = HistoryFragment.newInstance(historyList);
    fragmentTransaction.replace(R.id.fragment_container, fragment);
    fragmentTransaction.addToBackStack(null);
    fragmentTransaction.commit();
}
public void onPressCountbtn(View view) {
    Count++;
}

```

```

        tvCount.setText("Count: " + Count);
        historyList.add("Count: " + Count);
    }
}

```

XML CODE OF MAIN ACTIVITY:

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">
    <TextView
        android:id="@+id/text_view_count"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Count: 0"
        android:textSize="24sp" />
    <Button
        android:id="@+id/btn_count"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Count"
        android:onClick="onPressCountbtn"/>
    <Button
        android:id="@+id/btn_reset"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Reset"
        android:onClick="onPressResetbtn"/>
    <Button
        android:id="@+id/btn_view_history"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="View History"
        android:onClick="onPressHistorybtn"/>
    <FrameLayout
        android:id="@+id/fragment_container"

```

```
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_weight="1" />
</LinearLayout>
```

JAVA CODE OF History FRAGMENT:

```
package com.huzaiifashahbaz.tabesshfragmentapp;
import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import androidx.annotation.NonNull;
import androidx.annotation.Nullable;
import androidx.fragment.app.Fragment;
import java.util.ArrayList;
public class HistoryFragment extends Fragment {
    private static final String ARG_HISTORY = "history";
    private ArrayList<String> historyList;
    public static HistoryFragment newInstance(ArrayList<String>
history) {
        HistoryFragment fragment = new HistoryFragment();
        Bundle args = new Bundle();
        args.putStringArrayList(ARG_HISTORY, history);
        fragment.setArguments(args);
        return fragment;
    }
    @Nullable
    @Override
    public View onCreateView(@NonNull LayoutInflater inflater,
@Nullable ViewGroup container, @Nullable Bundle
savedInstanceState) {
        View view = inflater.inflate(R.layout.fragment_history, container,
false);
        ListView listView = view.findViewById(R.id.list_view_history);
```

```

        if (getArguments() != null) {
            historyList =
getArguments().getStringArrayList(ARG_HISTORY);
        }
        if (historyList != null) {
            ArrayAdapter<String> adapter = new
ArrayAdapter<>(getContext(), android.R.layout.simple_list_item_1,
historyList);
            listView.setAdapter(adapter);
        }
        return view;
    }
}

```

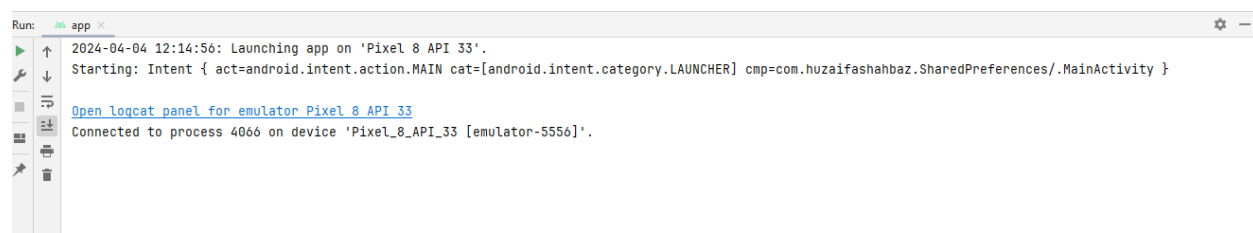
XML CODE OF History FRAGMENT:

```

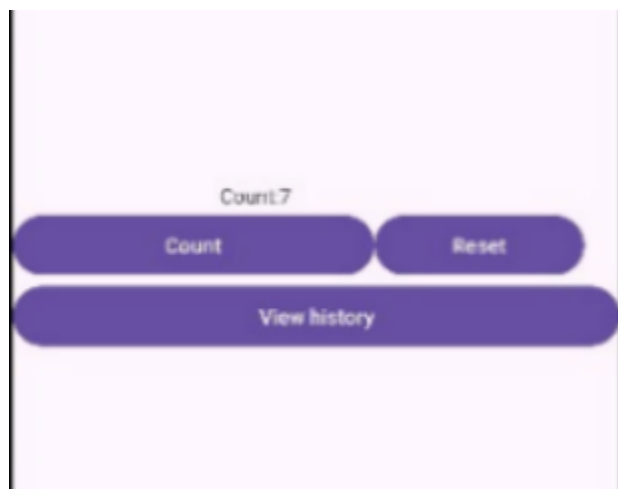
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">
    <ListView
        android:id="@+id/list_view_history"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />
</LinearLayout>

```

Results:



Output:



Tasbeeh History

Count 1: 5 Times

Saved at: 2023-12-04 22:36:27

Count 2: 9 Times

Saved at: 2023-12-04 22:43:44

Count 3: 6 Times

Saved at: 2023-12-04 22:36:26

Count 4: 33 Times

Saved at: 2023-12-04 22:36:48

Conclusion:

In conclusion we had learn how to make fragments app and also tabseeh fragment app by using history fragment in tasbeeh main activity.